

Natalie DeForest, PhD

Senior Bioinformatics Scientist & Computational Biologist

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Results-oriented scientist and leader with 8+ years of experience executing population genetics and multi-omics research initiatives. Expert at uncovering clinically actionable genomic insights via robust bioinformatics pipelines and machine learning frameworks. Proven ability to navigate and drive therapeutic innovation within both agile startups and large-scale organizations. Dedicated to building collaborative teams that turn multi-omics data into breakthroughs.

RELEVANT EXPERIENCE

Senior Scientist - Computational Genomics, Fauna Bio (Hybrid - Emeryville, CA / San Diego, CA)	2024 – 2026
As one of the first 20 employees: Identified translatable genetic targets for disease resistance via analysis of diverse multi-omics and genetic data (UK Biobank, MVP) to power Convergence, our AI-driven drug discovery platform. Led computational genomics efforts for target FN323G proposal with Eli Lilly; 323G approved as collaboration target in Feb 2026.	
Postdoctoral Scholar, Majithia Laboratory, School of Medicine, UC San Diego (La Jolla, CA)	2023 – 2024
Graduate Student Researcher, Majithia Laboratory, School of Medicine, UC San Diego (La Jolla, CA)	2018 – 2023
Integrated high-throughput genomic screens, diverse omics datasets, and large-scale genetic analyses to prioritize novel therapeutic targets for prevalent metabolic disorders (diabetes, cardiovascular disease, neurodegeneration, inflammation, cancer).	
Research Intern, Gilead Sciences (Foster City, CA)	Summer 2018
Research Intern, Cytokinetics (South San Francisco, CA)	Summer 2017
Undergraduate Honors Researcher, UC Davis Dept. of Chemistry (Davis, CA)	2016 – 2018
Clinical Data Intern, Pharmacyclics, an AbbVie Company (Sunnyvale, CA)	Summer 2016

SKILLS

- **Multi-omics and next generation sequencing (NGS) analysis** (bulk/single-cell RNA-seq (STAR, kallisto, DESeq2, edgeR, Seurat), samtools, ATAC/CLIP/ChIP-seq)
- **Statistical / population genetics analyses** (GWAS (Plink, Hail, REGENIE), omicQTLs, fine-mapping, colocalization, single variant/gene burden testing, Mendelian Randomization, PheWAS)
- **Scalable pipeline/workflow development** (Nextflow) and **relational database management** (SQL, Postgres)
- **Mining and analyzing relevant biological databases** (GTEx, ENCODE, Ensembl, GEO, SRA, ClinVar, 1000 Genomes, gnomAD) and **multi-modal clinical health data** (blood tests, biomarkers, patient histories)
- **R, Python, cloud-based computing** (AWS EC2/S3, GCP), and **data/code best practices** (Git version control)
- Familiar with **machine learning frameworks** (Tensorflow, scikit-learn) and genomic applications

SELECTED PUBLICATIONS & PRESENTATIONS

- **DeForest N.** et al. FB-1083 demonstrates potent efficacy in preclinical models of Heart Failure with Preserved Ejection Fraction and Pulmonary Hypertension. *Circulation / AHA* **2025**. Nov. 3, 2025. [10.1161/circ.152.suppl_3.4369301](https://doi.org/10.1161/circ.152.suppl_3.4369301)
- Guruge D, Li MM, **DeForest N.** et al. Cross-Species Graph Neural Network for Translating Animal Disease Resistance to Drug Targets. *NeurIPS 2025 AI Virtual Cells*. ai4d3.github.io/2025/papers/19_Cross_Species_Graph_Neural_.pdf
- **DeForest N.** et al. Genome-wide discovery and integrative genomic characterization of insulin resistance loci using serum triglycerides to HDL-cholesterol ratio as a proxy. *Nature Communications*. Sept 14, 2024. [10.1038/s41467-024-52105-y](https://doi.org/10.1038/s41467-024-52105-y)
- **DeForest N.** et al. Human gain-of-function variants in HNF1A confer protection from diabetes but independently increase hepatic secretion of atherogenic lipoproteins. *Cell Genomics*. May 30, 2023. [10.1016/j.xgen.2023.100339](https://doi.org/10.1016/j.xgen.2023.100339).
- **DeForest N, Majithia AR.** Genetics of Type 2 Diabetes: Implications from Large-Scale Studies. *Current Diabetes Reports*. Mar 19, 2022. [10.1007/s11892-022-01462-3](https://doi.org/10.1007/s11892-022-01462-3).

EDUCATION

PhD	Biomedical Sciences, Specialization in Bioinformatics, University of California San Diego	2023
	Dissertation: “Leveraging Human Genetics and Functional Genomics to Investigate Insulin Resistance Disorders”	
BS	Pharmaceutical Chemistry, Minor in Bioinformatics, University of California Davis	2018
	<i>summa cum laude</i> , Major/Minor GPA: 3.9	

CERTIFICATIONS / AWARDS

- **Machine Learning Specialization**, DeepLearning.AI + Stanford University, Coursera
 - **T32 National Research Award**, National Institute for General Medical Sciences (NIGMS)
- Dec 2024
2020-2023